

Explanation of Revisions

1 Summary of Major Changes

We have substantially revised the manuscript in response to all comments from the meta-reviewer (MR) and three reviewers (R1, R2, R3) from the previous ARR cycle. Below we provide an overview of changes followed by a point-by-point response.

Overview of major revisions:

1. **Backbone upgrade:** Replaced BiLSTM-CRF + GloVe with **RoBERTa-CRF and BERT-CRF** under the BOND distant supervision framework. All experiments and results now exclusively use transformer backbones. The BiLSTM-CRF text has been completely removed from the manuscript.
2. **Standard scheduler baselines:** Added Step Decay, Cosine Annealing, and ReduceLROnPlateau as comparison points. Results show these provide only marginal gains ($\pm 0.004 F_1$) over the static baseline, clarifying that the scheduling problem is non-trivial.
3. **Algorithm expansion:** Added GRPO (Group Relative Policy Optimization) as a sixth algorithm, representing recent advances in critic-free RL.
4. **Numerical results table:** Replaced all small, unreadable figures with a structured results table (Table 1) reporting test F_1 across all datasets, methods, and backbones, with 3-seed mean $\pm$ std.
5. **1-epoch vs. 2-epoch ablation:** A dedicated ablation study quantifying how per-trial epoch budgets affect each method differently.
6. **DQN search space documented:** Full specification of the 20-bin log-uniform action space, 6-dimensional state, epsilon-greedy schedule, and network architecture.
7. **Training cost analysis:** Per-method wall-clock timing on two GPU configurations (RTX 4060 vs. RTX 5090), with discussion of practical feasibility.
8. **Trade-off analysis:** Expanded comparative analysis section (Section 5) discussing BO vs. RL across sample efficiency, interpretability, convergence behavior, stability, and computational overhead.
9. **Writing and formatting:** Complete rewrite for professional tone; all grammatical errors and informal phrasing corrected; references verified and expanded; ACL-ARR formatting applied; anonymized submission; Limitations and Ethical Considerations sections added.
10. **Reproducibility:** Anonymous code package containing the full experiment framework, configuration files, and training scripts. Hyperparameter search spaces, dataset statistics, and implementation details provided in appendices.

2 Response to Meta-Reviewer (Overall Assessment: 2)

We thank the meta-reviewer for the constructive summary and clear guidance on the seven suggested revisions.

MR Revision 1: Clarify experimental setup consistently

Previous issue: The paper was unclear about whether experiments used BiLSTM-CRF+GloVe or BOND with RoBERTa, causing central confusion.

Revision: We have completely removed all BiLSTM-CRF content. The revised manuscript uses **RoBERTa-CRF and BERT-CRF exclusively** under the BOND framework (Section 4.1). We explicitly state the model configuration for every result in Table 1 (rows labeled RoBERTa-CRF and BERT-CRF). The BOND framework description specifies the exact pretrained checkpoint (roberta-base, bert-base-cased), the CRF layer configuration, and the self-training pro-

cedure for final models.

MR Revision 2: Stronger and more current baselines

Previous issue: Benefits of RL/BO scheduling shown only against a weak setting without transformer baselines or standard schedulers.

Revision: We now compare against: (1) **standard LR schedulers**—Step Decay, Cosine Annealing, ReduceLROnPlateau—as direct baselines for the scheduling problem (Table 1); (2) **two transformer backbones**—RoBERTa-CRF and BERT-CRF—as baseline architectures; and (3) **published SOTA methods**—BOND, SALO, RoSTER—in a comparison table (Table 2). These additions make the relative contribution of BO/RL scheduling clear: standard schedulers help marginally; backbone choice matters more than scheduling; and scheduling provides gains orthogonal to label denoising.

MR Revision 3: Readable tables with mean and standard deviation

Previous issue: Results were in unreadable figures only.

Revision: The primary results are now presented in **Table 1** (test F_1 scores, entity-level micro-averaged) with clear row/column labels. We report the best-trial test F_1 from the optimizer search phase, with 3-seed means (seeds 42, 123, 456) where compute permits (CoNLL-2003 experiments are ongoing, marked [TBD]). Optimizer trajectory figures are moved to the appendix (Section 9) with descriptive captions and enlarged for readability.

MR Revision 4: DQN search/action space details

Previous issue: DQN’s poor performance may depend on how discrete choices are defined.

Revision: Section 3.4 now fully specifies the DQN setup: (i) 20 **log-uniform bins** for the learning rate in $[10^{-5}, 10^{-2}]$; (ii) 6-dimensional **state space** (epoch index, validation F_1 , training loss, and their first differences); (iii) **epsilon-greedy exploration** decaying from 1.0 to 0.05; (iv) **network architecture** (2-layer MLP, 128 hidden units); (v) **replay buffer** size and target network update frequency. The complete hyperparameter search space is in Appendix A.

MR Revision 5: Explain BO vs. RL trade-offs beyond “X is better than Y”

Previous issue: Analysis was shallow—saying DQN is “very bad” without explaining why.

Revision: Section 5 (Comparative Analysis) now provides a structured trade-off discussion covering: (i) **sample efficiency**—GP-TS needs 10–20 trials, RL needs 20–50+; (ii) **interpretability**—GP surrogate provides uncertainty quantification and visualizable surface, RL policy networks are black-box; (iii) **convergence behavior**—TD3/SAC show earlier gains on cleaner datasets, GP-TS catches up on noisier ones; (iv) **stability**—GP-TS has lower variance across seeds than RL methods; (v) **training cost**—quantified in Table 3; and (vi) **scalability**—RL scales naturally to multi-dimensional action spaces, GP suffers from the curse of dimensionality. We also explain DQN’s underperformance: the discrete action space with only 20 bins lacks the granularity needed for fine-grained LR scheduling, and epsilon-greedy exploration leads to inefficient sampling compared to GP-TS’s uncertainty-aware sampling.

MR Revision 6: Writing and formatting revision

Previous issue: Informal phrasing, grammar issues, missing references, hard-to-read figures.

Revision: The entire manuscript has been professionally rewritten. Specific fixes: (i) all grammatical errors identified by reviewers (lines 47, 244, 434, 437, 452, 541 of the original) have been corrected in the corresponding sections of the revised manuscript; (ii) the informal “The DQN performance very bad” and “it can coverage” have been replaced with analytical descriptions of DQN results in context; (iii) the missing reference “(?)” has been corrected; (iv) all four reference entries with “Various” as placeholder authors have been verified and fixed; (v) the paper now uses ACL-ARR formatting with proper anonymization; (vi) figures have been enlarged with readable axis labels.

MR Revision 7: Strengthen reproducibility

Previous issue: No code, scripts, or protocol provided.

Revision: We provide an **anonymous code package** containing: (i) `experiment_runner.py`—the experiment orchestrator with artifact validation and cleanup policy; (ii) `optimizer_runtime.py`—shared optimizer utilities with trial management;

(iii) `train_final_models.py`—10-epoch final model training from best optimizer configurations; (iv) configuration YAML files for all experimental matrices; (v) the full hyperparameter search spaces (Appendix A); (vi) dataset statistics and the BOND-based preprocessing protocol (Appendix C); (vii) implementation details including random seeds, hardware specifications, and software versions (Appendix B).

3 Response to Reviewer 1

R1 Summary: “Low baseline: BiLSTM-CRF with GloVe is outdated... No comparison against standard LR schedulers. Some figures are unreadable. Writing quality is poor. Citation appears as ‘(?)’.”

We appreciate the reviewer’s candid assessment. The specific issues have been addressed as follows:

- **BiLSTM-CRF baseline (R1’s main concern):** Completely replaced. All experiments now use RoBERTa-CRF and BERT-CRF. The old BiLSTM-CRF content (Section 4, old lines 195–284) has been removed entirely. We further added BERT-CRF generalization results to demonstrate that backbone choice is itself a critical factor.
- **Standard LR schedulers:** Added Step Decay, Cosine Annealing, and ReduceLROnPlateau (Table 1). The finding that these provide only $\pm 0.004 F_1$ over the static baseline is, we believe, itself informative.
- **Unreadable figures:** Removed as primary presentation format. Results now in structured tables. Remaining figures (Appendix, Section 9) have been enlarged with proper axis labels.
- **Writing quality:** The entire manuscript has been professionally edited. All informal phrasing (“The DQN performance very bad,” “it can converge”) has been removed and replaced with analytical text. The entire Section 4 (old) has been restructured.
- **Missing citation “(?)”:** Corrected. The reference at the corresponding location has been verified and properly formatted.

4 Response to Reviewer 2

R2 Summary: “Weak baseline with BiLSTM-CRF. Results in figures are almost invisible. Need early stopping, transformer baselines, tables, recent references, multiple seeds. Writing and punctuation need fixing.”

We address each concern:

- **BiLSTM-CRF baseline:** Removed. Replaced with RoBERTa-CRF and BERT-CRF. We thank the reviewer for this suggestion, which significantly strengthened the paper.
- **Invisible figures:** Replaced with a comprehensive results table (Table 1). Figures moved to appendix with enlarged rendering.
- **Early stopping:** Added to all training runs. The revised training pipeline implements early stopping (patience=3 on validation F_1). This is now documented in Section 4.1 and Appendix B.
- **Transformer baselines:** Both RoBERTa-CRF and BERT-CRF now serve as the primary backbones. BERT-CRF results (Table 1, bottom rows) demonstrate substantially stronger performance, confirming the reviewer’s insight that backbone choice is critical.
- **Numerical tables:** Table 1 now provides the complete results in tabular format with clear labeling.
- **Recent references:** The reference list has been expanded with recent 2023–2025 publications including ULTHO (ICLR 2025), GPBT (2024), OpenAutoNLU (2025), Context-Aware AutoML (2025), RoSTER (EMNLP 2023), and AutoDistil (2023).
- **Multiple seeds:** Results now span three seeds (42, 123, 456). The 3-seed protocol is documented in Section 4.2.
- **Writing:** All punctuation, comma, and full-stop issues have been fixed throughout the manuscript.

5 Response to Reviewer 3

R3 Summary: “DQN search space not described. Analysis shallow. BO vs. RL trade-offs missing. Training time not reported. Citation error and typos.”

We address each concern:

- **DQN search space:** Fully specified in Section 3.4: 20 log-uniform bins for learning rate in $[10^{-5}, 10^{-2}]$, 6-dimensional state space, epsilon-greedy schedule, 2-layer MLP architecture. Complete hyperparameter details in Appendix A.
- **Shallow analysis:** The problematic text (“The DQN performance is very bad, and the SAC/TRPO seems similar to TD3 but a little weaker”) has been removed. Section 5 now provides a structured, evidence-based comparative analysis of all six methods across multiple dimensions (sample efficiency, interpretability, convergence, stability, cost, scalability). DQN’s under-

performance is explained analytically: the 20-bin discretization loses information relative to continuous action spaces, and epsilon-greedy exploration is less efficient than GP-TS’s uncertainty-guided sampling.

- **BO vs. RL trade-offs:** Now the central focus of Section 5. We discuss: GP-TS treats each trial independently (contextual bandit), benefiting from sample efficiency but missing temporal patterns; RL methods learn state-conditioned policies that exploit training dynamics but require multi-epoch trial budgets. We also quantify the trade-off in Table 3 (training cost) and discuss practical scenarios where each paradigm is preferable.
- **Training time:** Table 3 reports per-cell wall-clock time for each method on RTX 4060 (8GB) and RTX 5090 (32GB). SAC requires $\sim 2.5\times$ more time than GP-TS/TD3 due to 50-trial budgets. We also note that BO/RL optimization is a one-time cost.
- **Citation error and typos:** Fixed. The “(?)” citation has been corrected. All typos in the original text have been corrected throughout.

6 Additional Improvements Over the Previous Submission

Beyond reviewer-requested changes:

- **GRPO algorithm:** Added as a sixth method, representing recent advances in critic-free RL (Group Relative Policy Optimization).
- **SOTA comparison table (Table 2):** Positions our results among published methods (BOND, SALO, RoSTER), clarifying that scheduling is complementary to denoising.
- **Best-configuration analysis (Section 4.4):** GP-TS posterior analysis reveals that noisier datasets reward more aggressive learning rates and longer warmup—an actionable finding for practitioners.
- **Code release:** Anonymous zip package with full experiment framework, validated for zero sensitive information leakage.
- **Limitations section (Section 7):** Discusses computational cost, incomplete CoNLL-2003 optimizer results, scope limitation to LR scheduling, and RoBERTa vs. BERT gap.
- **Ethical Considerations (Section 8):** Addresses GPU energy consumption, dataset provenance, and potential NER bias.
- **ACL-ARR formatting:** Fully compliant with the ACL style guide and ARR anonymity require-

ments.

We believe these revisions address all concerns raised by the meta-reviewer and all three reviewers, significantly strengthening the paper’s experimental grounding, presentation clarity, analytical depth, and reproducibility. We thank all reviewers for their constructive feedback.